



Alex Towell

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Alex Towell

PhD Student | Mechanistic Interpretability | AI Safety

About Me Computer Science PhD student with dual master's degrees (CS & Mathematics/Statistics) seeking to understand how neural networks represent and process information. Recent work applies network analysis to AI conversations, revealing hidden cognitive structure—directly analogous to circuit analysis in neural networks. I value reproducibility, systematic evaluation, and honest uncertainty quantification.

Publications

- **A. Towell**, J. Matta, “Cognitive MRI of AI Conversations: Analyzing AI Interactions through Semantic Embedding Networks,” *Complex Networks & Their Applications (CNA)*, 2025
- H. Fujinoki, **A. Towell**, V.A. Thota, “Preventing Ransomware Damages using In-Operation Off-Site Backup to Achieve a 10^{-8} False-Negative Miss-Detection Rate,” *ICCCI*, 2025
- **A. Towell**, “Reliability Estimation in Series Systems,” Master’s Project, SIUE, 2023
- **A. Towell**, H. Fujinoki, “Estimating How Confidential Encrypted Searches Are,” *IEEE CloudCom*, 2016

Education

2024 - Present, Southern Illinois University

PhD in Computer Science Focus: AI Safety, Interpretability, Network Analysis. Advisor: Dr. Hiroshi Fujinoki

2021 - 2023, SIUE

MS in Mathematics (GPA: 3.9/4.0) Concentration: Statistics and Operations Research

2012 - 2015, SIUE

MS in Computer Science (GPA: 4.0/4.0) Concentration: Artificial Intelligence. Outstanding Graduate Student Award.

2006 - 2010, SIUE

BS in Computer Science (GPA: 3.6/4.0)

Technical Skills

Languages

Python (primary), R, C/C++ , TypeScript, Bash, LaTeX

ML/DL

PyTorch, HuggingFace Transformers, NumPy, Pandas, NetworkX

Methods

Graph/network analysis, statistical inference, ablation studies



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Relevant Research & Projects

Cognitive MRI of AI Conversations (CNA 2025)

Applied complex network analysis to 449 ChatGPT conversations to reveal latent cognitive structure. Identified heterogeneous topology and three distinct bridge types. Comprehensive ablation study across 63 parameter configurations.

elasticsearch-lm: LLM Fine-Tuning for Query Generation

Fine-tuning TinyLlama to translate natural language to Elasticsearch queries. Synthetic data generation, evaluation on populated databases. PyPI published, 36+ GitHub stars.

Ransomware Detection

Detection system achieving 10^{-8} false-negative rate using Bloom filters with rigorous statistical evaluation.

Fisher Flow

Natural gradient descent using Fisher information matrix and information geometry.

Oblivious Computing

Bernoulli types for principled approximation; algebraic cipher types for homomorphic computation; encrypted search with information-theoretic bounds.

Why Mechanistic Interpretability

- **Cognitive MRI work:** Network analysis methods to reveal hidden structure in AI conversations—directly analogous to circuit analysis in neural networks
- **Mathematical foundations:** MS in Mathematics with focus on probability, statistics, and analysis
- **Systematic evaluation:** All my research emphasizes ablation studies, uncertainty quantification, and reproducible methodology
- **Implementation depth:** FemtoGrad and similar projects demonstrate understanding systems at the implementation level

Academic Experience

2024–Present, Research Assistant, SIU

- Network analysis of AI systems with Dr. John Matta (Cognitive MRI paper)
- Cybersecurity research with Dr. Hiroshi Fujinoki on adversarial robustness
- Lead developer for IDOT D8 traffic monitoring infrastructure

2012–2015, Research/Teaching Assistant, SIUE

- AI/ML research on privacy-preserving search and encrypted computation
- Taught undergraduate computer science laboratories